1. Define Classification.

Classification is a supervised learning technique used to assign input data to predefined categories or classes.

- It learns from labelled training examples and predicts the class of new data.
- The output is a discrete value, such as Yes/No, True/False, or Spam/Not Spam.
- Example: Classifying an email as Spam or Not Spam based on its contents.

2. Define Regression.

Regression is a supervised learning technique used to predict continuous numerical values based on input attributes.

- It identifies the relationship between input variables and the output.
- In linear regression, the relationship is represented as:

$$y = w_1 x + w_0$$

where w_1 is the weight and w_0 is the intercept.

- Example: Predicting the price of a house based on its floor area.

3. Define Overfitting.

Overfitting occurs when a machine learning model learns the training data too closely, including random noise and unnecessary details, instead of identifying the actual pattern.

- The model performs well on training data but poorly on unseen data.
- It commonly occurs when the model is too complex.
- Example: A decision tree learns irrelevant features such as the color of a die to predict whether it will show a six.

Overfitting can be reduced through pruning, regularization, and appropriate model selection.

4. Explain Decision Trees with an Example.

A decision tree is a supervised learning method that represents decisions using a tree-like structure of attribute tests and outcomes.

- Root node: Represents the first attribute tested.
- Internal nodes: Represent tests on input attributes.
- Branches: Represent possible values of the attributes.

- Leaf nodes: Represent final decisions or classifications.

Example: Restaurant Problem

A decision tree determines whether a customer should wait for a table based on the number of patrons.

Simplified restaurant decision tree

- If Patrons = None, the decision is No.
- If Patrons = Some, the decision is Yes.
- If Patrons = Full, additional attributes such as waiting time and hunger are checked.

5. Describe Theory of Learning with the Restaurant Problem.

The theory of learning explains how a learning algorithm can develop hypotheses that predict unseen examples accurately.

- It uses PAC (Probably Approximately Correct) learning, which aims to produce a hypothesis with low error and high probability.
- In the restaurant problem, the agent learns whether a customer will wait for a table using attributes such as Patrons, Hungry, and Fri/Sat.
- A decision list can represent the learned hypothesis:

$$WillWait \Leftrightarrow (Patrons = Some) \vee (Patrons = Full \wedge Fri/Sat)$$

Example: The customer waits if there are some patrons, or if the restaurant is full on Friday or Saturday.

PAC learning indicates that sufficient training examples help the model make approximately correct predictions with high probability.

6. Explain Statistical Learning with the Candy Problem.

Statistical learning is a learning approach that uses probability and observed data to make predictions under uncertainty.

Candy Problem:

- A bag contains two flavors of candy: Cherry and Lime.
- There are five possible types of candy bags, represented by hypotheses h_1, h_2, h_3, h_4, h_5.

- The agent observes the flavors of candies and updates the probability of each hypothesis using Bayes' rule.

$$P(h_i \mid D) = \frac{P(D \mid h_i)P(h_i)}{P(D)}$$

where D represents the observed candy flavors.

Example: If the first ten candies are Lime, the agent assigns a higher probability to the all-Lime bag and predicts that the next candy will also be Lime.

7. Describe Explanation-Based Learning to Extract General Rules.

Explanation-Based Learning (EBL) is a learning technique that uses prior knowledge and logical reasoning to derive general rules from specific examples.

Steps involved:

1. Construct an explanation: Explain why the given example or solution is correct using existing knowledge.
2. Generalize the explanation: Identify the essential conditions that make the solution valid.
3. Form a general rule: Create a reusable rule that can solve similar problems.

Example:

Given the expression:

$$1 \times (0 + X) = X$$

Using the rules $0 + Y = Y$ and $1 \times Y = Y$, the system derives a general simplification rule.

Thus, EBL improves problem-solving efficiency by avoiding repeated reasoning.

8. Define Data Matrix.

A data matrix is a rectangular arrangement of data in rows and columns used to represent a dataset in machine learning.

- Each row represents an observation or training example.
- Each column represents an attribute or feature.
- For m observations and n attributes, the data matrix has dimensions

Example:

Studen	Study	Mark

t	Hours	s
S1	2	50
S2	4	70
S3	6	85

Here, each row represents a student and each data column represents an attribute.

This is a supplementary standard definition; the supplied notes do not explicitly define a data matrix.

9. Define Density Estimation.

Density estimation is an unsupervised statistical learning technique used to estimate the probability distribution of data from observed samples.

- It identifies how data values are distributed.
- It estimates regions where observations are more or less likely to occur.
- Density estimation may use parametric or non-parametric methods.

Example: Estimating the probability distribution of students' examination marks to identify commonly occurring score ranges.

This is a supplementary standard definition; the supplied notes do not explicitly define density estimation.

10. Explain Q-Learning.

Q-Learning is a model-free reinforcement learning algorithm in which an agent learns the utility of performing an action in a particular state.

- It maintains a Q-value Q(s,a) for every state-action pair.
- The Q-value is updated based on the reward received and the expected future reward.
- The agent selects actions with higher Q-values to maximize total reward.

Q-Learning update rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where is the learning rate, is the discount factor, and r is the reward.

Example: A robot learns the best path to a destination by receiving positive rewards for reaching the goal and negative rewards for undesirable movements.

11. Define Utility Function.

A utility function is a mathematical function that represents the desirability or usefulness of a state to an intelligent agent.

- It assigns a numerical value to each state.
- Higher utility indicates a more desirable state.
- The agent selects actions that maximize expected utility.

Example: In a grid-world environment, reaching a goal state gives a reward of +1, entering a dangerous state gives −1, and each intermediate step may have a reward of −0.04.

Thus, the utility function helps the agent make rational decisions.

12. Define Learning Curve.

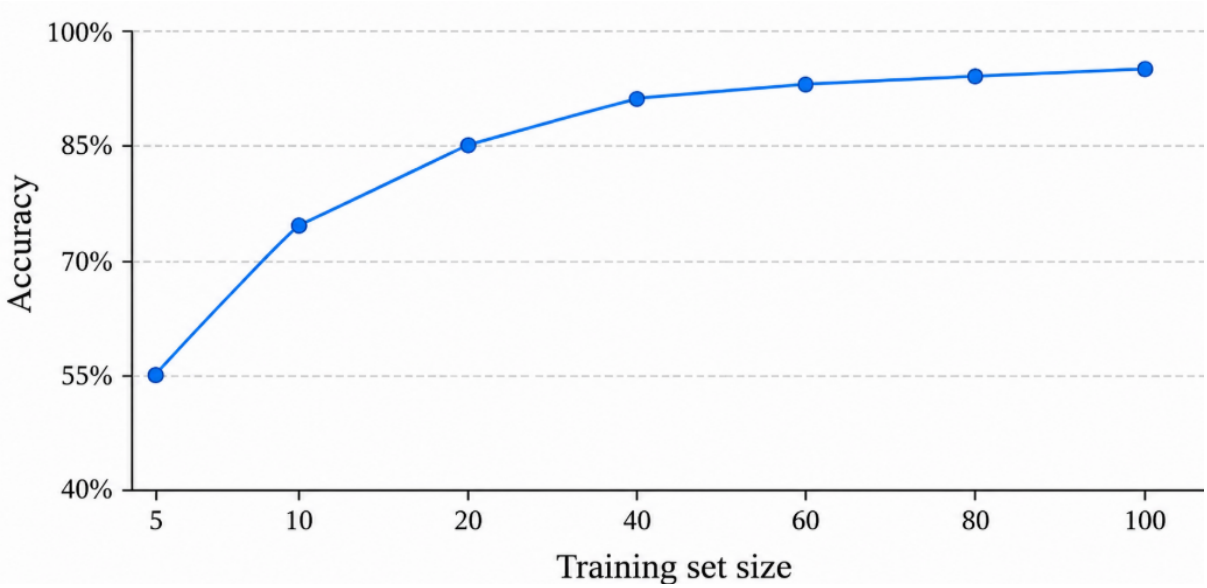
A learning curve is a graphical representation that shows how the performance of a machine learning algorithm changes as the number of training examples increases.

- The X-axis represents the training set size.
- The Y-axis represents accuracy or another performance measure.
- It helps evaluate how well the model learns and generalizes to new data.

Example: In the restaurant decision-tree problem, prediction accuracy increases with more training examples and approaches approximately 95% in the example given in the notes.

Illustrative learning curve

Conceptual example; not the exact measurements from the notes.



13. Define Explanation-Based Learning.

Explanation-Based Learning (EBL) is an analytical learning method that uses existing domain knowledge to explain a training example and generalize it into a reusable rule.

- It uses logical reasoning rather than relying only on repeated observations.
- It identifies the essential conditions responsible for a successful solution.
- It improves efficiency by storing generalized rules for future use.

Example: After understanding that the derivative of X^2 is $2X$, the system generalizes the rule:

$$\frac{d(u^2)}{du} = 2u$$

This rule can be applied to different variables.

14. Define the Term "Reward" in Reinforcement Learning.

A reward is a numerical feedback signal received by an agent from the environment after performing an action.

- A positive reward indicates a desirable outcome.
- A negative reward indicates an undesirable outcome.
- The objective of reinforcement learning is to learn a policy that maximizes the expected cumulative reward.

Example: In a chess game, winning may give a reward of +1, while losing may give a reward of −1.

Rewards help the agent identify beneficial actions and improve future decisions.

15. What Is Overfitting in Decision Trees?

Overfitting in decision trees occurs when the tree becomes excessively complex and learns noise or irrelevant patterns from the training data.

- It creates unnecessary branches to classify training examples.
- It achieves high training accuracy but poor prediction accuracy on unseen examples.
- It can be reduced by pruning branches that provide little or no useful information.

Example: A decision tree may use irrelevant features such as the color or weight of a fair die to predict whether it will show a six.

Prevention: Decision-tree pruning, statistical significance testing, and selecting an appropriate tree size.

16. Differentiate Between Positive and Negative Reinforcement.

Positive and negative reinforcement are methods used to encourage desirable behavior by providing or removing a consequence.

Positive Reinforcement	**Negative Reinforcement**
Involves adding a desirable stimulus after an action.	Involves removing an unpleasant stimulus after an action.
Increases the likelihood of repeating the behavior.	Also increases the likelihood of repeating the behavior.
Example: Giving a reward to a robot for reaching its destination.	Example: Removing an unpleasant warning sound when a robot performs the correct action.

Conclusion: Both positive and negative reinforcement aim to strengthen desirable behavior. Positive reinforcement adds a desirable stimulus, whereas negative reinforcement removes an unpleasant stimulus.